

Introduction

- Alport syndrome: loss of *Col4a3*, *a4*, or *a5* (1).
- Losing one gene blocks assembly — the glomerular basement membrane (GBM) weakens and fails (2).

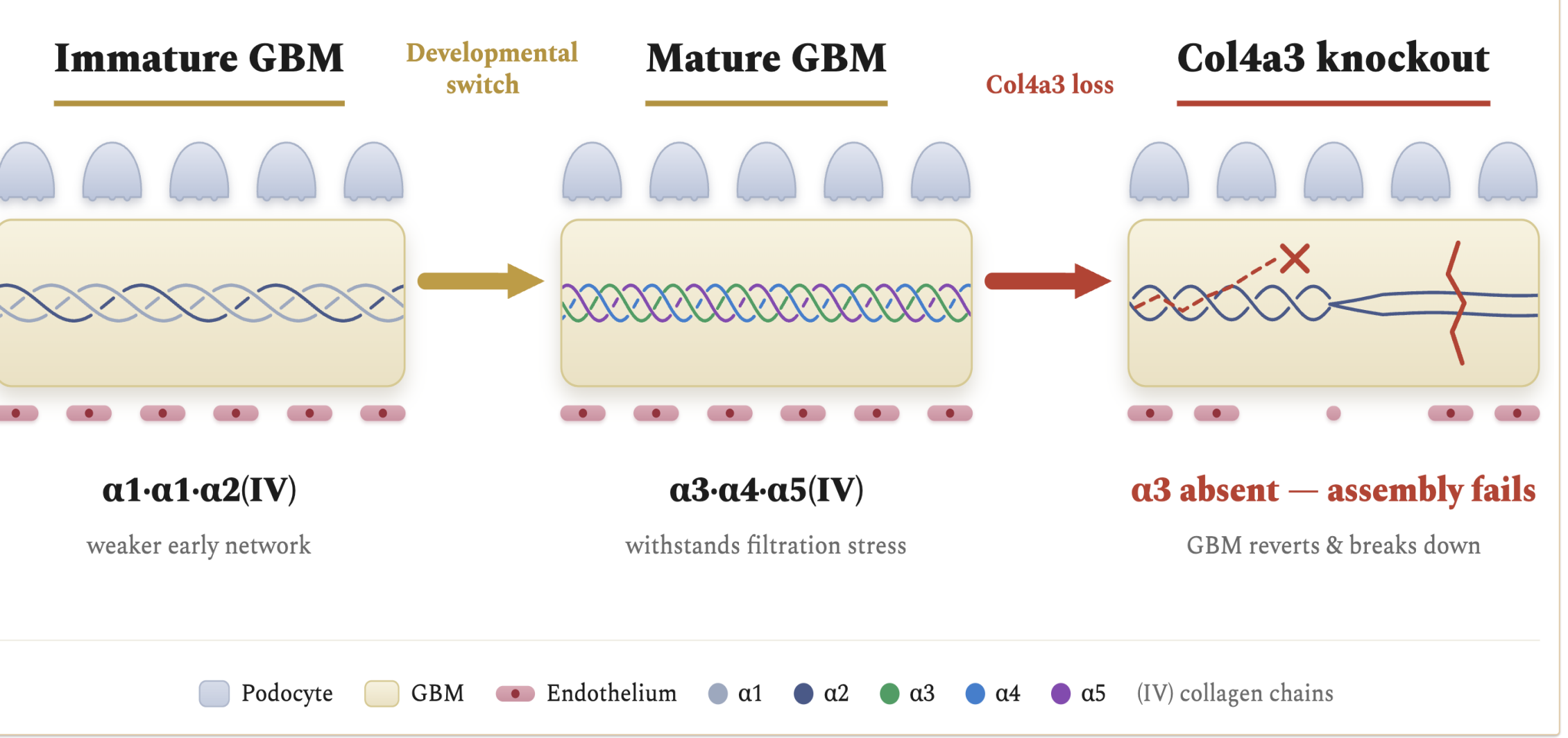


Fig 1. The collagen IV network switch in the GBM, and where *Col4a3* loss breaks it.

Hypothesis

The *Col4a3* knockout cell line would share similarities with knockout mouse tissue but not fully reflect the complex *in vivo* changes.

Methods

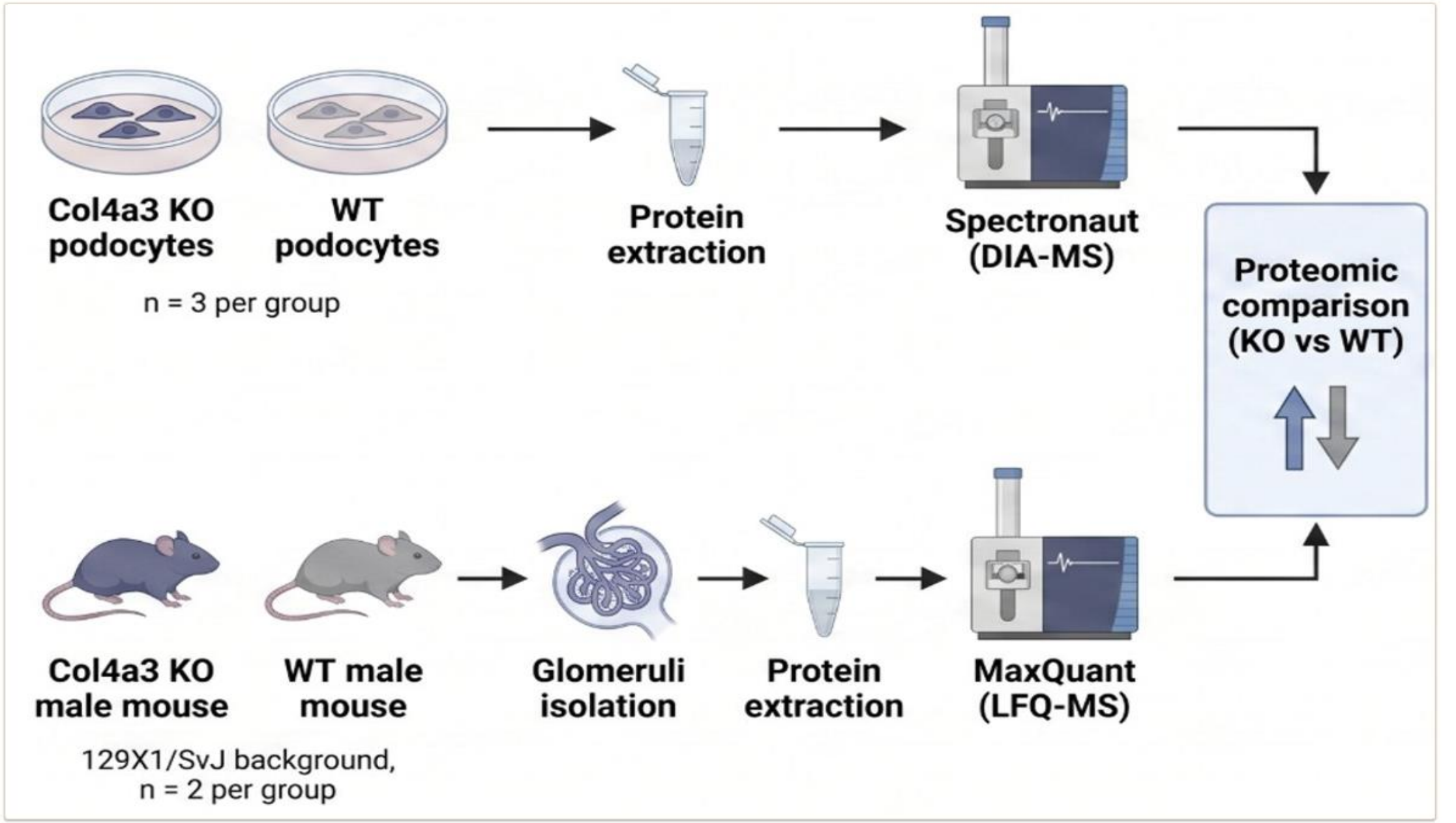


Fig 2. Knockout (KO) vs wild-type (WT) workflows: cell line and male mouse tissue.

Results

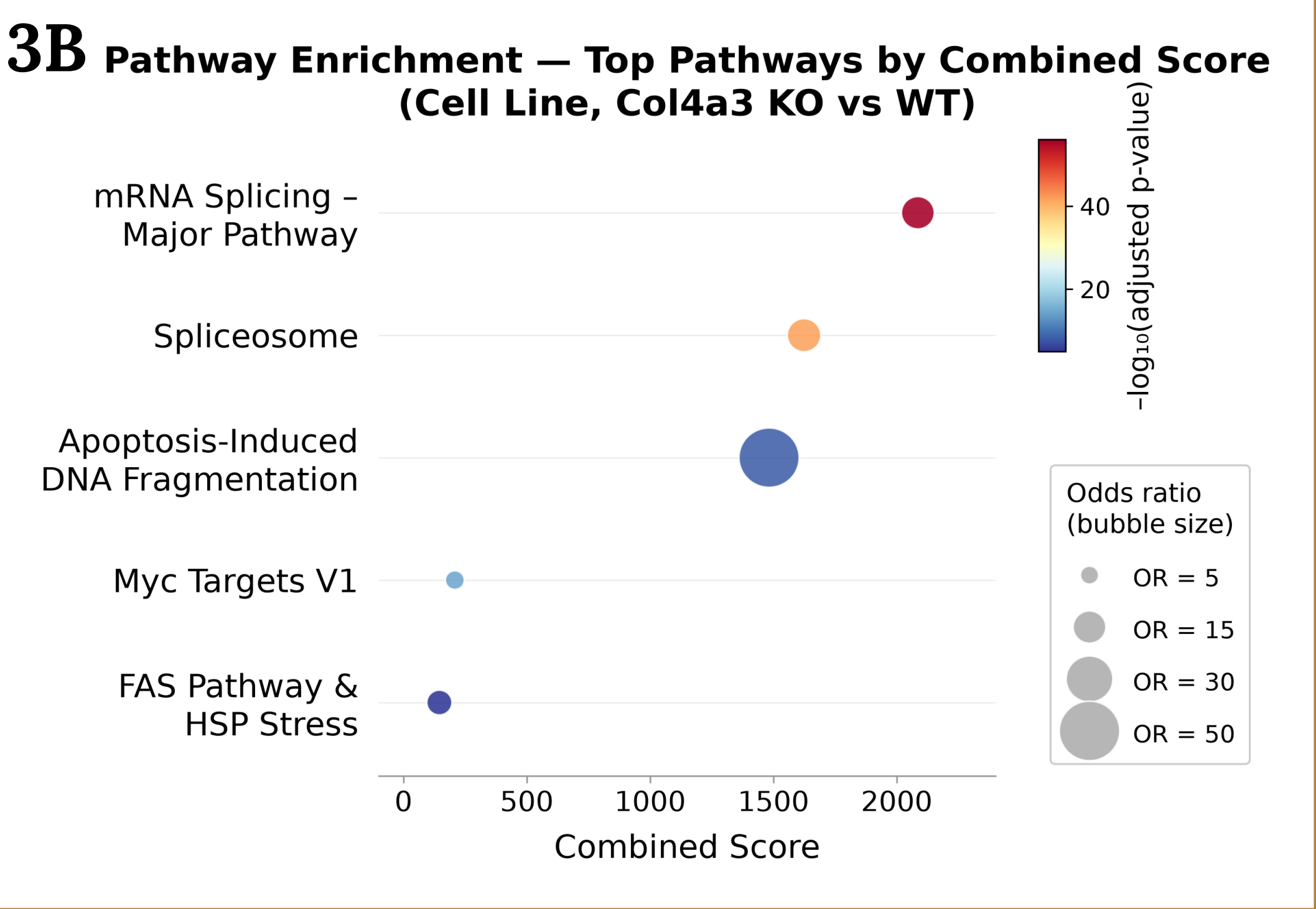
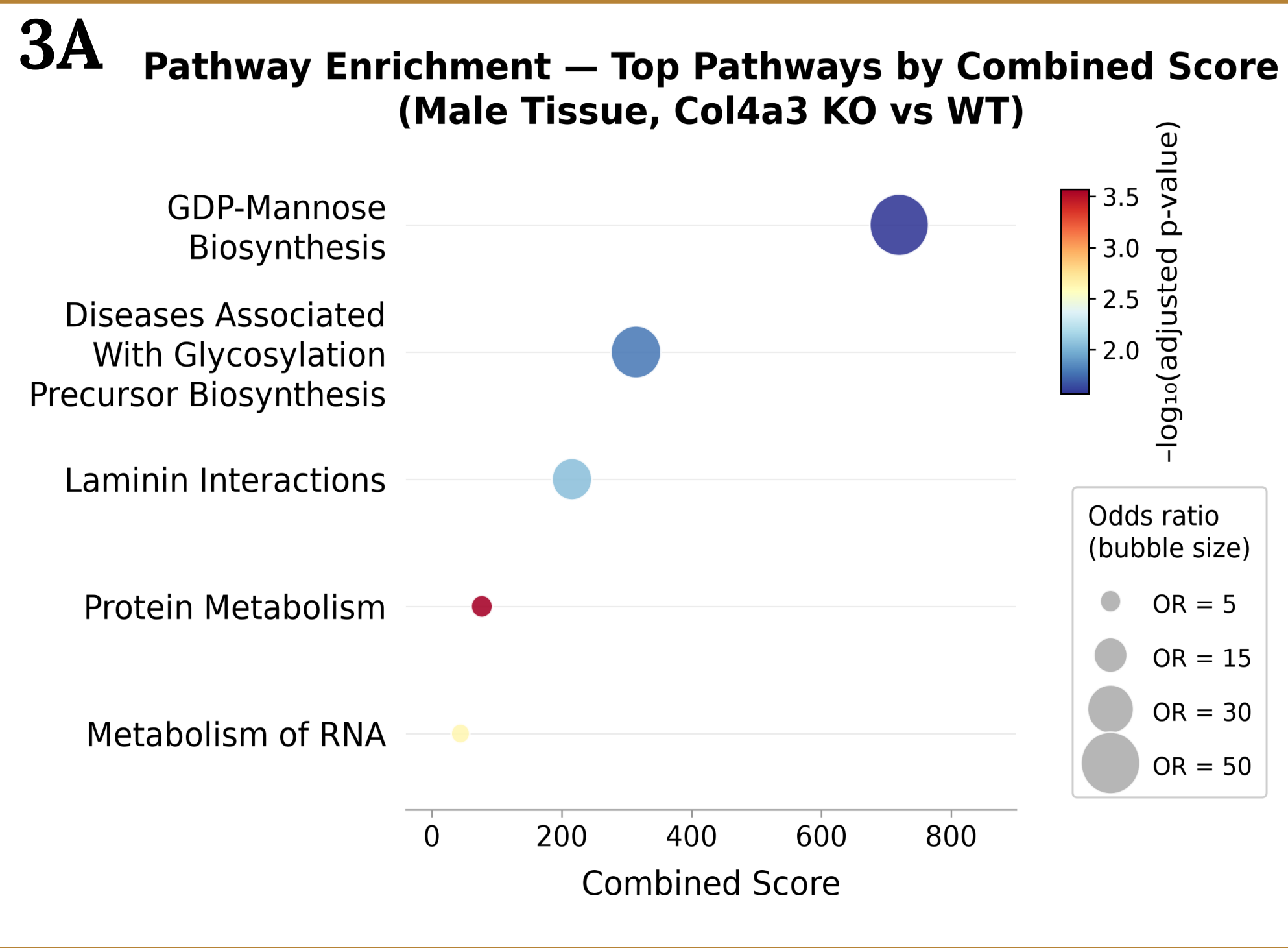


Fig 3. (A) Male tissue enrichment is led by glycosylation and laminin pathways.

(B) Cell line enrichment is dominated by mRNA splicing.

Fig 4. (A) Differentially expressed fibrosis-associated candidates (4)

(B) 3,414 proteins are shared between cell line and male tissue.

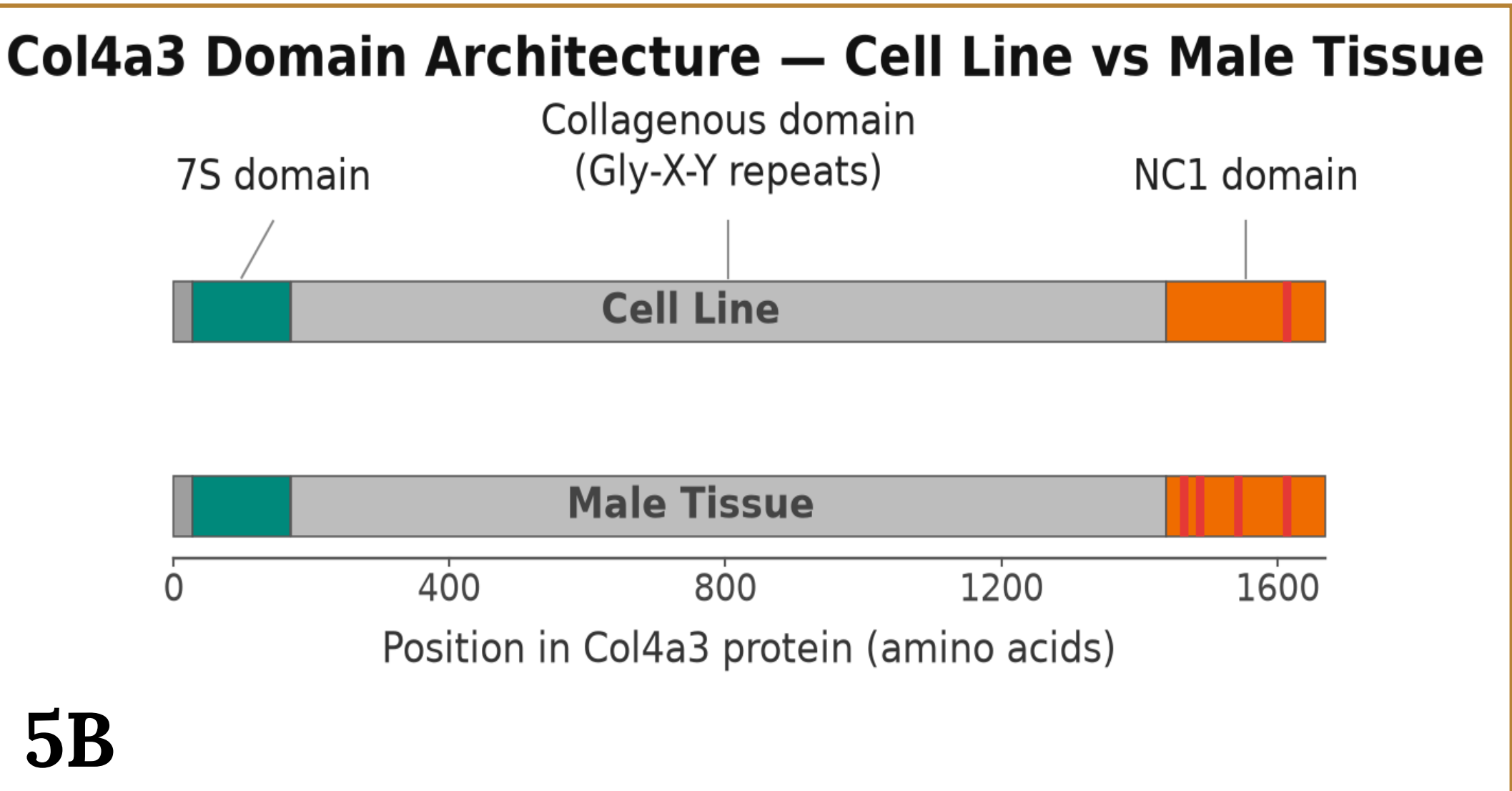
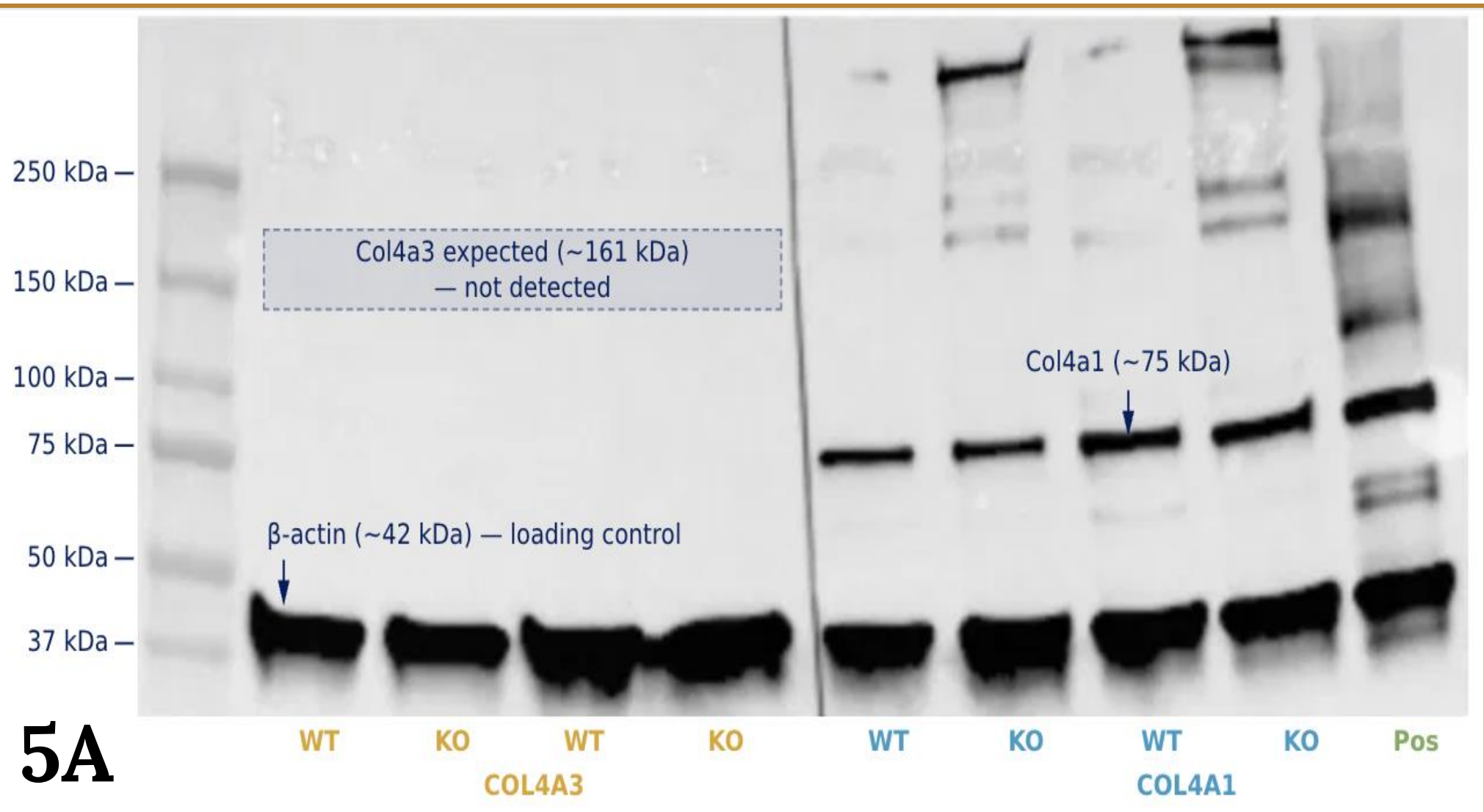
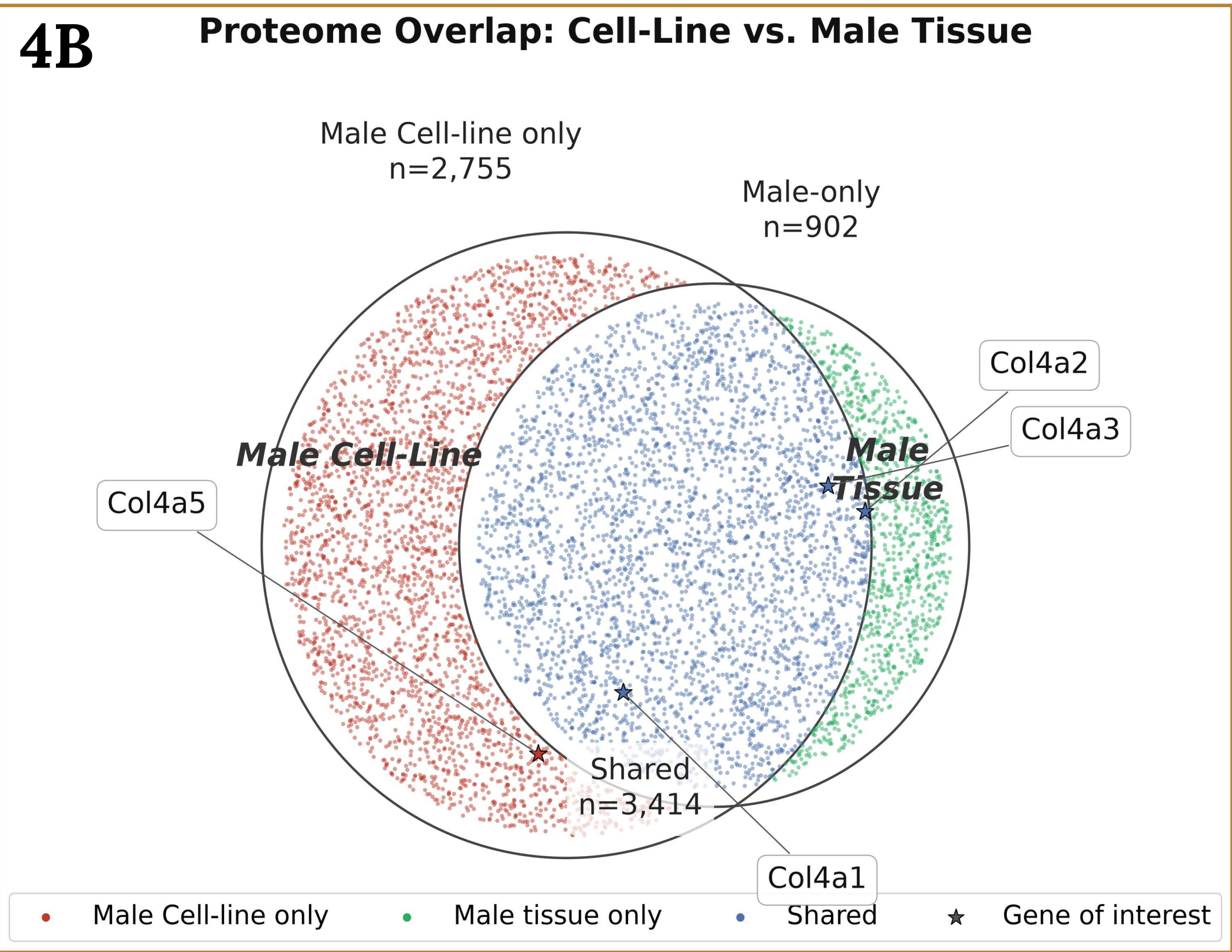
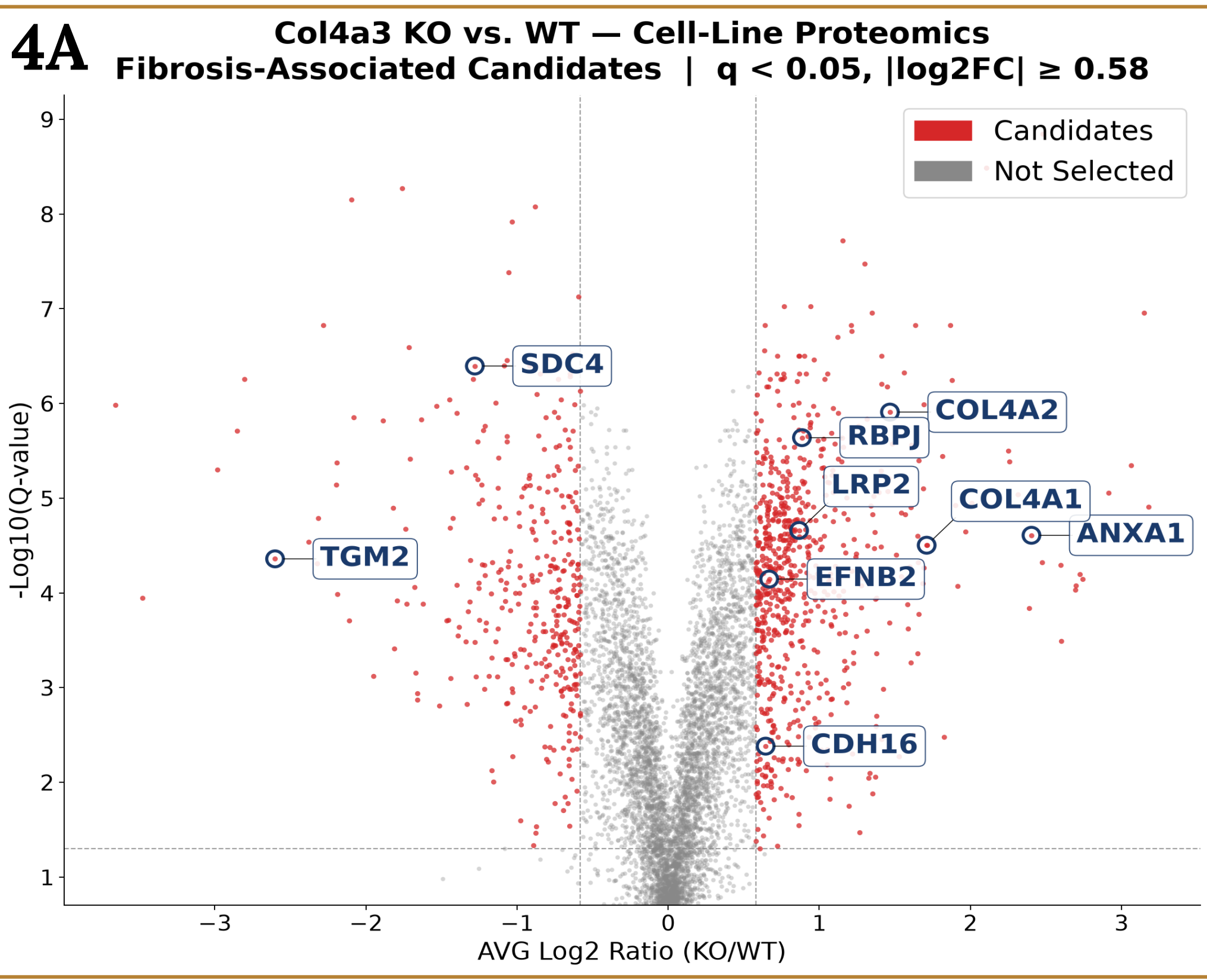


Fig 5. (A) Blot shows *Col4a1*, but no *Col4a3*, even in the WT for cell line.

(B) Mass spec maps *Col4a3* to NC1 domain in both models.

Conclusion

- Cell line confirms *Col4a3* loss and shares the RNA-processing / spliceosome signal with tissue.
- Only 4 of 23 shared fibrosis markers agree in direction across all datasets (4).
- Col4a1* and *Col4a2* rise in the cell line but fall in tissue.
- The cell line should not be used as the sole predictor of *in vivo* *Col4a3* knockout.

Limitations

- Mouse tissue: n = 2 per group and sampled, underpowered
- Cell line lacks GBM mechanical stress and multicellular context

References

- (1) De Gregorio et al. *Kidney Med.* 2023;5:100631.
- (2) De Gregorio et al. *Nat Rev Nephrol.* 2025;21:200-215.
- (3) Caparali et al. *JASN.* 2025;36:1176-1183.
- (4) Gu et al. *Renal fibrosis gene panel*, 2021.